RESTORATION ECOLOGY
The Journal of the Society for Ecological Restoration

SER SOCIETY FOR ECOLOGICAL RESTORATION

RESEARCH ARTICLE

Restored native prairie supports abundant and species-rich native bee communities on conventional farms

Caitlin Paterson[1], Karl Cottenie[1], Andrew S. MacDougall[1,2]

Native pollinators are increasingly needed on conventional farms yet rarely fostered via management. One solution is habitat restoration in marginal areas, but colonization may be constrained if resident pollinator richness is low or if restored areas fail to provide sufficient floral or nesting resources. We quantified restoration outcomes for native bees, and associated resources, on three conventional farms with forb-grass prairie plantings on marginal areas of varying sizes, in a heavily farmed region of central North America. We tested bee abundance and richness in restored prairie versus the dominant habitats of the region—crops, forest remnants, and edges of fields and roads. Restored prairie supported 2× more species (95 of 119 total species) and 3× more bees (72% of captured individuals) compared to the other cover types. All richness and abundance differences among habitat types were associated with higher floral resources in restored prairie. Thirty percent of the bee species were unique to prairie, consistent with long-distance dispersal but begging the question of origin given the absence of prairie regionally. Our results suggest that road and field edges may be the source, as these areas had more floral and nesting resources than forest or crop fields combined and supported 55% of all species despite covering only approximately 5% of the sampled farms. Habitat scarcity is not the only constraint on native bees in agricultural landscapes, with increasing concern over disease and chemicals. However, we observed that restored areas on marginal lands of conventional farms can support abundant and species-rich populations of native bees.

Key words: conventional farms, flowering resources, native pollinators, nesting resources, tallgrass prairie

Implications for Practice

- Restored high-quality habitat on "marginal" lands of conventional farms significantly increased the diversity and abundance of native bees, despite being mostly small and spatially isolated.
- Forty-two percent of the 119 species only occurred in restored habitat, in association with floral resources, representing all major bee functional groups including ground nesters.
- Other than restored habitat, edges of roads and fields supported the most species—far more than forest or crop fields—suggesting that they could serve as refugia for bees in the absence of high-quality restored prairie habitat.
- Although scarcity of high-quality habitat is not the only constraint on native bees in agricultural regions, restored areas on marginal land appear capable of rapidly building local populations.

Introduction

There is increasing concern over the potential declines of native pollinators on intensively managed agricultural landscapes (i.e. nonorganic conventional farms) (Kremen et al. 2002; Ponisio et al. 2016; Carvell et al. 2017; Kovács-Hostyánszki et al. 2017). This may be especially true where factors such as disease are affecting honey bees, such that native species can provide important alternatives for pollination services (Cox-Foster et al. 2007; Winfree et al. 2007; Garibaldi et al. 2013; Kennedy et al. 2013; Barral et al. 2015). Rarely, however, do conventional farms manage for the floral and nesting resources necessary to support native pollinators, creating a potential gap between the increasing need for services provided by native pollinators versus the inadvertent loss of habitat including reduced floral and nesting resources (Koh et al. 2016).

Considerable research on native pollinator diversity in agricultural landscapes has contrasted organic versus conventional farms, typically finding the latter to support significantly fewer and less diverse assemblages (Kremen et al. 2002; Morandin & Winston 2005; Schuler et al. 2005; Holzschuh et al. 2007). However, the distribution and abundance of remnant pollinator

Author contributions: CP, ASM developed the research and wrote the paper; CP conducted all field work including coordinating bee identification; KC assisted with statistical analysis and writing the paper.

[1]Department of Integrative Biology, University of Guelph, Guelph, Canada N1G 2W1
[2]Address correspondence to Andrew S. MacDougall, email asm@uoguelph.ca

doi: 10.1111/rec.12987
Supporting information at:
http://onlinelibrary.wiley.com/doi/10.1111/rec.12987/suppinfo

diversity on conventional farm landscapes tend to be less well described. Diversity in such landscapes could be chronically low given factors such as pesticide impacts and habitat loss (Kremen et al. 2002; Winfree et al. 2009; Carvell et al. 2017), although these impacts may differ among conventional farms depending on how each is managed. For those species that do persist, intensive agricultural management may more negatively affect some functional groups than others, such as ground nesters constrained by repeated soil cultivation (Kim et al. 2006). It is also unclear, with extensive habitat loss on agricultural landscapes, if and where persisting native pollinators reside, with the potential for crop fields, remnant forest, or the edges of fields or roads to provide suitable habitat even though the latter two cover an increasingly small percentage of farm area (Kerr & DeGuise 2004; Wood et al. 2015). Furthermore, given that these suitable areas are increasingly spatially restricted in intensively managed regions, remnant populations of pollinators could be isolated depending on whether the distance separating suitable remnant area exceeds the dispersal distance of persisting species (Carstensen et al. 2016).

Given the importance of habitat for pollinators, one management solution is restoration (Öckinger & Smith 2007; Hopwood 2008; Hannon & Sisk 2009; Winfree 2010; Morandin & Kremen 2013*a*, 2013*b*; Blaauw & Isaacs 2014; Scheper et al. 2015; Williams et al. 2015; Wood et al. 2015; Ponisio et al. 2016; Carvell et al. 2017; Kaiser-Bunbury et al. 2017). But success may depend on the current diversity, distribution, and abundance of native bee populations (Kovács-Hostyánszki et al. 2017)—colonization may be limited to those species present within a given farm. Success may also vary by the suitability, diversity, and amount of floral and nesting resources in restored areas (Morandin & Kremen 2013*a*) versus those found in seminatural habitats such as road and field edges—restored areas may have limited impacts on many bee species if the resources provided are not substantially greater than other areas (Scheper et al. 2015; Wood et al. 2015).

Here, we test the impacts of restoring resource-rich tallgrass prairie habitat on a major group of pollinators—native bees—in an intensively managed agricultural region of central North America (Norfolk County, Canada). These restored areas occurred on formerly cropped marginal lands of three farms, with planted native tallgrass prairie and savanna (prairie with sparse tree cover) that can be rich in floral and nesting resources for bees (Hines & Hendrix 2005). Native tallgrass prairie and savanna formerly covered approximately 1,000–2,000 km^2 of Ontario as recently as the early 1800s; it now occurs in approximately 1% of that range with most existing prairie being restored starting in the 1990s (Riley 2013). For our study, prairie restoration was a farmer-led initiative (ALUS Canada, http://alus.ca) targeting marginal crop fields that sought to provide various ecosystem services including pollinators, while minimizing impacts on crop yields (Robertson & Swinton 2005; Jack et al. 2008).

Using spatially and temporally stratified sampling of native bees over one growing season, we tested their relative species richness and abundance among four cover types on each farm (restored prairie and oak savanna [hereafter "prairie" for simplicity]), crop fields including orchards on one farm, forest patches, and edge areas. Our analyses investigated (1) how bee abundance and richness varied by the interacting effects of season (June, July, August), farm (three farms), and cover type (planted prairie, edge, forest, crop), (2) which types of resources were associated with bee composition by cover type and season (e.g. flower color, nesting material) including assessing how the spatial distribution of resources affected native bee occurrences (using redundancy analysis [RDA] and PCA), and (3) the availability of floral and nesting resources among cover types by sampling period. Given the rarity of high-quality grassland habitat regionally, the young ages of the restored prairie (<10 years), and their relatively small sizes and isolation on farms, we predicted that the restored prairies would not differ from other cover types in bee richness and composition. Furthermore, given that restoration success of prairie can sometimes be variable (Biondini 2007; MacDougall et al. 2008), we also tested the differences in floral and nesting resource availability among restored versus non-restored habitat, especially whether forest and edge areas might provide similar levels of pollinator resources (Carvell et al. 2007; Wood et al. 2015).

Methods

Study Area

We worked in Norfolk Country, Ontario, along the northern shoreline of Lake Erie in central North America (Paterson 2014). The climate of the region is moderated by its proximity to Lake Erie, which is the shallowest and warmest of the Great Lakes. The regional mean annual precipitation is 1,100 mm, with growing season temperatures averaging 18.5°C (https://weather.gc.ca/mainmenu/weather_menu_e.html).

Our study region, prior to the 1800s, was a mixture of tallgrass prairie, oak savanna, and deciduous "Carolinian" forest. It is now intensively managed for a range of crops including soybean and corn, grains, vegetables, fruit and berry orchards, tobacco, ginseng, and peanuts; many farms also include small patches of deciduous forest. Crop fields and small forest patches cover 89% of the region (A. Dolezal, pers. comm.), with the remainder as wetlands, riparian areas, and settlement. There is no remaining unbroken prairie.

We worked on three farms that are part of a regional network of farmers (ALUS Canada) restoring native tallgrass prairie on marginal areas of farms for ecosystem services including pollinators, soil carbon, cattle forage, and the capture nutrient runoff from crop fields (http://alus.ca/newacreproject/). Plantings were species-rich mixtures of native perennial grasses (C_4 species—typically dominated by *Andropogon gerardii*, *Sorghastrum nutans*) and forbs (typically including species with high nectar rewards—*Verbena stricta*, *Asclepias tuberosa*, *Monarda fistulosa*). Prairie plants were added as seed to plowed areas that previously supported crops, with the prairies of this study planted 5–8 years prior. These planted areas were "marginal" in the sense of requiring the highest inputs of fertilizer, irrigation, or both to produce profitable crop yields, usually on sandy and infertile post-glacial "eolian sand" soils common in the region. Most ALUS farms are conventionally managed with fertilization and various chemical, plowing, and crop rotations that are increasingly dominated by corn and soybeans.

The three farms were chosen based on two criteria. First, they were separated by a minimum of 25 km, to ensure independence of bee communities, while facilitating coordinated sampling of native bees within the same time and weather conditions. Second, they had various combinations of our targeted cover types: crop or orchards, edge areas, forest patches, and restored prairie including prairie "oak savanna" (Fig. S1). Farm 1 covered 13 ha and was dominated by corn fields (60% cover), with planted prairie (25%), forest (10%), and edge habitat (5%). Farm 2 covered 20.3 ha and was dominated by planted prairie and oak savanna (60%), with berry production (20%), forest (15%), and edge (5%). Farm 3 covered 49.5 ha and was dominated by cropping (70% corn, soybeans, squash, pumpkins), with planted prairie (15%), forest (10%), and edge (5%).

Sampling. Our sampling design centered on having identical collection effort per farm in each time period (Krebs 1998), with forty 16-m^2 plots per farm. Given that the relative percentages of the four cover types differed among farms (e.g. prairie sizes ranged from 15 to 60% of total sampling area), we sought to reduce sampling bias by proportionately assigning the 40 plots per farm differently, so that sampling effort per cover type was similar (e.g. prairies occupying more of the total farm area were allocated more of the 40 plots—see Fig. S1). In total for the 120 plots on the three farms, there were 50 prairie, 31 crop, 20 forest, and 19 edge plots. Within each cover type, plots were randomly placed to capture variability in environmental conditions and "distance" (e.g. locations in crop fields both near and far away from boundaries with prairie [represented by the black dots in Fig. 1A–1C]).

We sampled native bees and associated floral and nesting resource availability in identical locations at three times: June, July, and August 2013. The multiple sampling was used to capture expected seasonal changes in both native bee and floral resources between late spring, midsummer, and late summer. The initial sampling occurred in June, rather than earlier, because cold weather in April–May 2013 delayed native bee emergence and flowering of pollinator-friendly plants.

Within each 16-m^2 plot, we sampled both bees and the abundance and availability of nesting and floral resources. Native bees were captured in pan traps painted with blue or yellow fluorescent Kryolan spray paint (two of each color; four pans per plot) and sweep netting using 40-cm-diameter fine mesh nets (Spafford & Lortie 2013). The pan traps, placed in the plots over an 8-hour period starting before 9 am, were filled with five drops of dish soap per 1,000 mL of water. The traps were placed on the ground in spring, and then on elevated narrow platforms just above the canopy in summer to maintain visibility for foraging bees. Sweep netting was conducted from 10:00 to 14:00 to maximize catch rate when bees are at their peak flight abundance (Droege et al. 2010). Sweeping occurred across the entire 16-m^2 plot, targeting the vegetation canopy where most flowers were found. It is important to combine pan traps and sweep netting, especially since the former can sometimes be taxonomically biased for certain native bee species (Spafford & Lortie 2013).

Specimens were immediately stored in an ice-chest and later frozen. Each bee was identified visually using keys for Apoidea found on discoverlife.org and Marshall (2007) (Tables S1–S3). Difficult-to-identify specimens were sent for DNA barcoding at the Canadian Centre for DNA Barcoding in Guelph, Ontario. Records are stored in the Barcode of Life Data System, including photographs of specimens (see "ASGCB–Grassland Community Barcoding" file at http://www.boldsystems.org).

Nesting was estimated using percent coverage of five substrate types (bare ground, mud, pithy stems/twigs, woody debris, and existing cavities >2 mm which may have been created by other insects in the ground or in dead plant material and might be suitable for bees) to cover the range of materials and nesting strategies utilized by native bees (Packer et al. 2007; The Xerces Society 2011). The availability of floral resources was visually estimated as percent cover of flowers per plot in 1% increments, for white, blue/purple, and yellow flowering species, presence of pollen and nectar availability (scored as the presence of flowers known to produce nectar and those with visible pollen), and the richness of those plants currently flowering (at the lowest taxonomic level possible) (Kremen et al. 2004).

Analysis

We conducted three types of analyses. First, we used generalized linear modeling (GLM) with Poisson distribution to test how our count data of species richness and abundance per plot varied by the interacting effects of season (June, July, August), farm (three farms), and cover type (planted prairie, edge, forest, crop). Poisson-based GLMs were needed because our data were zero-inflated, with multiple crop and forest plots in particular having no captures especially in July and August. We did not rarefy our richness data, because the species accumulation curves reached clear asymptotes for each of the four cover types (Fig. S2). Second, we used analysis of variance (ANOVA), forward step-wise regressions, and repeated measures ANOVA with post hoc Tukey tests to determine how cover type and season affected the availability of resources (e.g. flower color, nesting material). All noncount data were log +1 transformed to improve normality. All analyses were conducted with JMP (version 14), with farms treated as a random factor (SAS Institute Inc., Cary, NC).

Finally, we used a range of complementary RDA and principle components (PCA) analyses to test the influence of environmental cues—specifically nesting and floral resources by cover type—on bee abundance and richness (Tables S4–S6; Figs. 1 & S2; Cottenie 2005; Pinto & MacDougall 2010; Germain et al. 2013). Given that bees are assumed to disperse wide distances to forage, there could be strong environmental signals even though high-quality habitat can cover relatively small areas of some of the farms (e.g. native bee richness and abundance reflect resource availability, regardless of the isolation of high-quality habitat on a farm). Alternatively, patches of high-quality habitat surrounded by unsuitable habitat such as crop monocultures

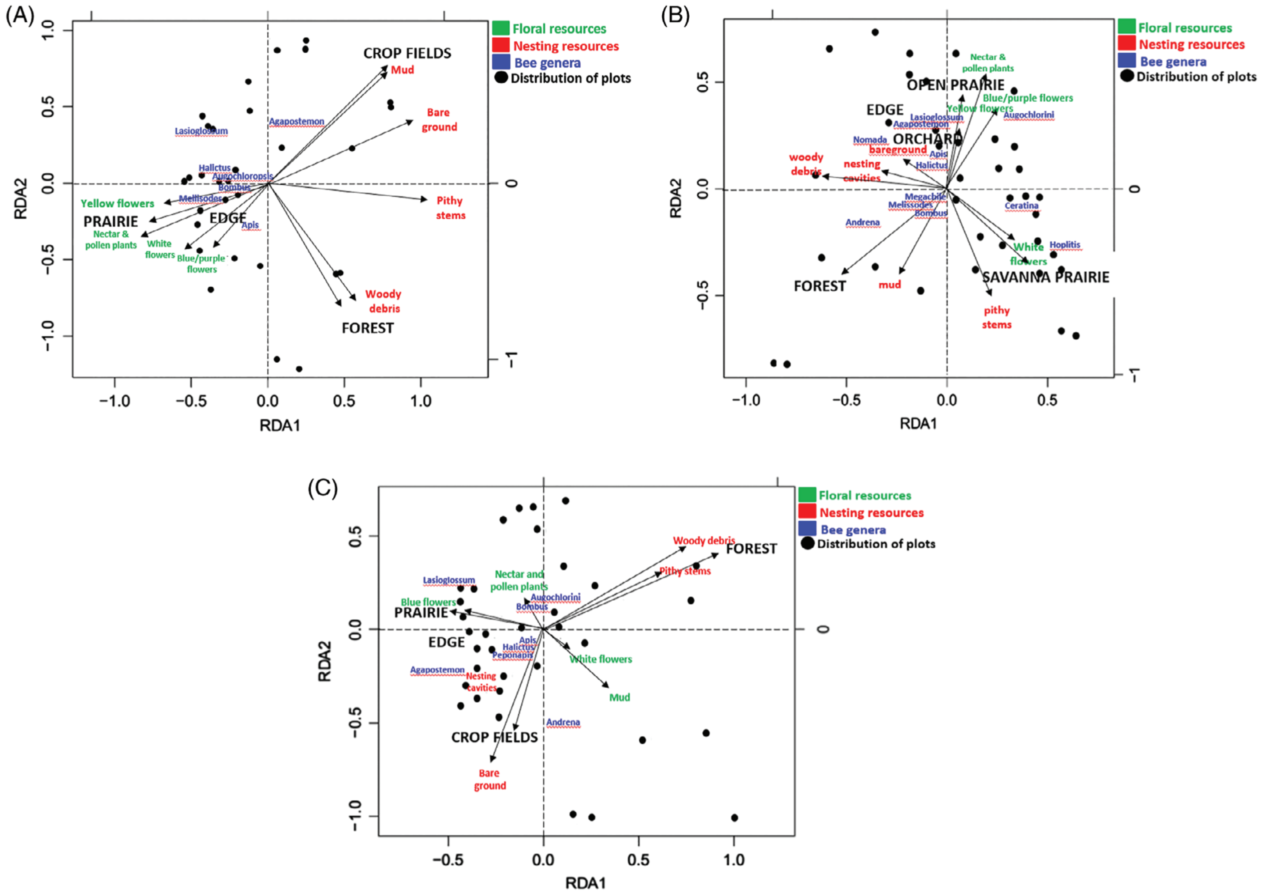

Figure 1. RDA plots for the three study areas ([A] Farm 1, [B] Farm 2, [C] Farm 3) showing the association among the four major cover types (prairie, edge, crop fields, and forest), the distribution of floral (green text) and nesting resources (red text), and the distribution of the surveyed plots in relation to the cover types on each farm (solid black dots). The genera of native bee most spatially separated between the two RDA axes are shown in blue text; the remaining bee genera were tightly congregated at the intersection of the RDA1 and RDA2 axes and are not shown. Adjusted r^2 values: Farm 1: $r^2 = 0.146$; Farm 2: $r^2 = 0.237$; Farm 3: $r^2 = 0.164$ (see further details in Appendix S1).

may have reduced richness and abundance, because they are isolated enough that bees are less likely to reach them despite their dispersal capabilities. We used a distance-based RDA (db-RDA) with Jaccard's dissimilarity distances to compute the variation in the dependent pollinator community matrix explained by the relative influence of the environmental and distance matrices (see Appendix S1 for complete details). This method combines RDA and PCA, with the freedom to choose between a wide variety of distance measures including those that are non-Euclidean (Anderson et al. 2011; Borcard et al. 2011). We used Jaccard's dissimilarity distances because this binary presence-absence matrix allows us to look at the composition of a community when there is a large range of abundances and zero entries in the species data matrix (Borcard et al. 2011).

Results

We captured 119 bee species, of the 420 native bee species known for our region of Ontario (Packer et al. 2007). Most individuals were captured in blue (58%) and yellow (39%) pans, with the remainder captured by sweep netting. Sweep netting captured numerous insects, including many native pollinating flies, but relatively few native bees compared to the pan trapping. The captured species represented 22 bee genera totaling 1,243 individuals (Tables S1–S3). We barcoded 190 individuals to confirm identity of species identified manually, or for species difficult to identify (e.g. *Lasioglossum*, *Andrena*, *Megachile,* and *Nomada*). Thirty-four of the 190 barcoded samples were contaminated with *Wolbachia*, which prevented identification of the native bee to species, while suggesting infection may be common among native bees in our region. The most abundant families were Halictidae (58% of all individuals), Apidae (20%), Megachilidae (16%), Andrenidae (4%), and Collitalae (0.4%). The most abundant genera were *Lasioglossum*, *Agapostemon*, *Augochlorella*, *Ceratina*, and *Hoplitis*, with *Hoplitis pilosifrons* and *Augochlorella aurata* the most abundant species (Table S1). The rarest genera were *Calliopsis*, *Colletes*, *Dieunomia*, *Svastra*, and *Xylocopa*. Thirty-six species were collected only once in the entire survey, with most captured in prairie and, to a lesser degree, edge habitat (Table S2). Most of the common

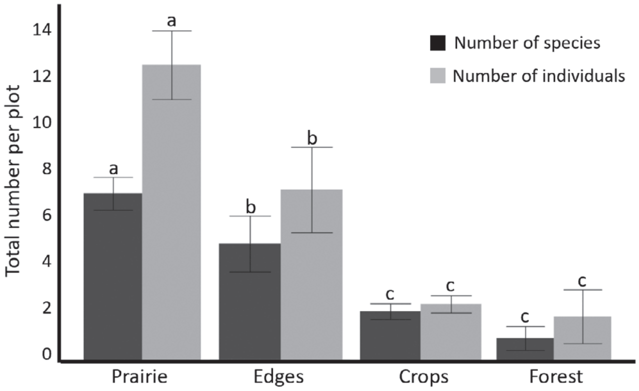

Figure 2. Total number per plot of native bee species and bee individuals in the four cover types, pooled for our three study areas over the three sampling periods. Error bars are ±1 SE. Letters indicate significant differences among number of species and number of individuals separately, derived from Tukey's post hoc comparisons.

bee genera are classified as floral generalists (e.g. *Lasioglossum*, *Andrena*, and *Megachile*—Packer et al. 2007; The Xerces Society 2011).

The species richness and total abundance of native bees per plot were explained by higher order interactions among cover type, farm, and month of sampling (Tables S4–S6; Figs. 1 & 2). These interactions involved the significant impact of prairie versus the other three covers on both plot-level richness and abundance (abundance:likelihood ratio [LR] = 178, $df = 3$; $p < 0.0001$; richness LR = 91, $df = 3$; $p < 0.0001$), the higher numbers of species on Farm 2 with its larger extent of prairie compared to Farms 1 and 3 (Fig. S1; abundance LR = 42.3, $df = 6$; $p < 0.0001$; richness LR = 30, $p = 6$; $p < 0.0001$), but also wider seasonal fluctuations in bees on Farm 2 with significantly higher numbers per plot in June versus the rest of the summer (Fig. 3; abundance LR = 178, $df = 4$; $p < 0.0001$; richness LR = 90, $df = 4$; $p < 0.0001$).

Most species and individuals were found in planted prairie compared to edge areas, crop fields, and forest—these patterns were observed on all farms and during each of the three sampling periods (Figs. 1–3). Prairie habitats supported 80% of the bee species, including 30% that only occurred in these areas (Table S3). Prairie habitat had the greatest range of flower color, the most abundant and diverse floral resources (pollen- and nectar-bearing plant species), and greater duration of flower availability from June to August compared to crop fields and forest (but not edges) (Tables S4–S6; Figs. 1–5). As well, pithy stems and twigs were abundant in prairie habitat with bare ground to a lesser extent (Tables S4–S6; Figs. 1 & 5). Stepwise analysis of floral versus nesting resources failed to detect any influence of the latter on bee richness, with total floral resources per plot being the only factor with significant explanatory power ($F_{[9,118]} = 76.8$; $p < 0.0001$). That being said, the RDA analyses suggest that the strong influence of floral resources on bee richness derives from a subset of highly abundant genera that were disproportionately captured in prairie (Fig. 1A–1C), especially *Lasioglossum*, *Halictus*, *Agapostemon*, *Apis*, and *Auglochlorini*. Many genera, however, did not show this strong bias toward prairie, and at least one genera—*Andrena*—was found more in crop fields and forest (Fig. 1B & 1C). These results were generally consistent among the three farms. We did not directly test seasonal availability of bees versus flowers in prairie, but they appear to be negatively correlated with the highest number of flowers per plot in August (Fig. 4A) and the highest numbers of bees per plot in June (Fig. 4B).

Although prairie supported greater bee richness, bee abundance, and floral resources, edge areas still supported 55% of all captured species despite only covering approximately 5% of all areas sampled (range: 3.2–11.6% of total farm area sampled). Ten species were exclusive to this cover type (Table S3). If we simply examine occurrences of species per plot, given that the coverage of edge areas was relatively small compared to the other cover types, then the richness per unit area (but not abundance) was higher in edge areas than prairie (Table S3). Many of the edge plots had the same characteristics associated with prairie plots, especially total availability of floral resources, the seasonal availability of floral resources, and nesting resources (Figs. 1–5, S2). Indeed, the RDA analyses tended to group edge plots in close association with prairie plots (Fig. 1). The availability of flowering plants was generally lower in edge than prairie in terms of number of species, percentage cover of floral resources, and range of flower colors (Tables S4–S6). There were plant compositional differences between prairie and edge areas, with planted prairie dominated by native plants, while edge habitat contained regional widespread non-native "old-field" species including clovers, vetches, and thistles (all of which provide floral resources for pollinators). The predominance of blue and purple flowers in edge plots in late summer (Fig. 3) reflects the relative high abundance of non-native vetches (e.g. *Vicia cracca*) and thistles (*Circium* spp.).

Crop fields supported 28% of all captured species. Bare ground was typically abundant (Tables S4–S6; Figs. 1 & 5), although it was unclear whether this would provide successful nesting habitat to bees due to periodic tilling. Most bees in crop fields were captured near prairie and edge areas (data not shown), with almost half of the sampling capturing zero individuals in one or more of the survey periods.

Forest supported 18% of captured species. Forest was rich in nesting material but generally lacked flowers during the sampling periods (Figs. 4 & 5) including in spring when ephemeral understory plants can sometimes be abundant. Woody debris, pithy stems and twigs, and some bare ground were abundant in forested areas (Tables S4–S6). Three-quarters of all sampling in the forested areas of the three farms captured zero individuals.

Discussion

We examined the impacts of habitat type on native bees on conventional farms, focusing on farms with restored high-resource prairie habitat on marginal areas. A question of interest is whether these plantings, even over small areas, could positively affect the richness and abundance of native bees versus other cover types on the farms. We found this to be the case—restored

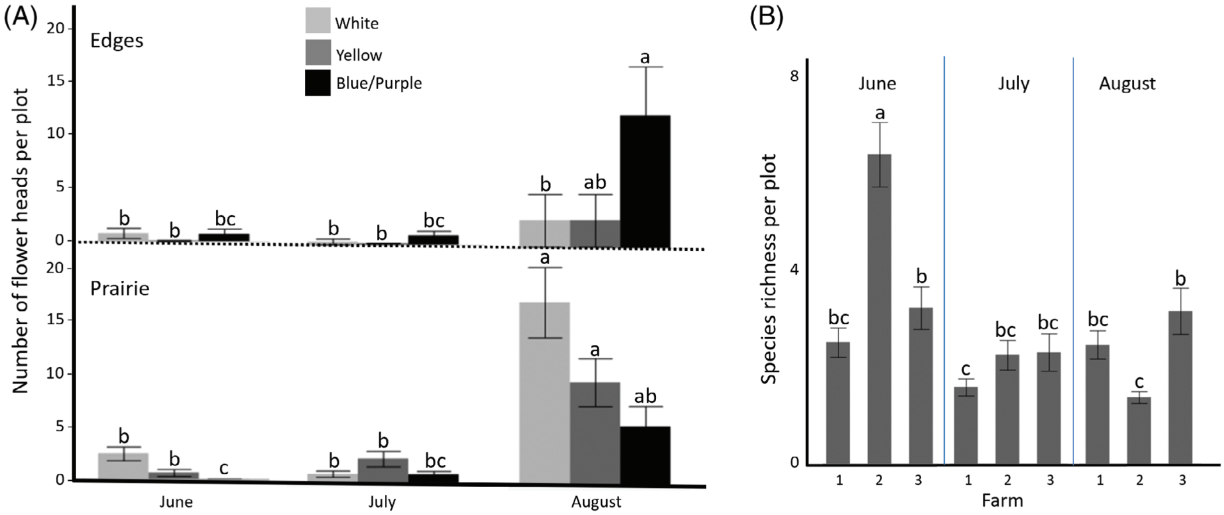

Figure 3. Seasonal variation in floral resources and bee capture by farm. (A) Total number of flower heads per plot, for the three color categories, by season for edge areas and prairie. The other cover types—crop fields and forest—generally lacked floral resources in any of the sampling periods. Letters indicate Tukey's post hoc comparisons for each of the three colors separately, between the two cover types. (B) Average number of native bee species captured per plot by season for the three farms. Data are pooled among the four cover types of prairie, crop field, forest, and edge habitat for each farm. Farms 1 and 3—both farms with corn-soy crop fields—have similar levels of pollinator richness by season. Farm 2 has an unexplained decline in native bees from June to August, despite floral resources in prairie and edge areas peaking in August on all farms. Error bars are ±1 SE. Given the significant interaction between farm and season ($F_{[4, 359]} = 18.8$; $p = 0.001$), letters indicate Tukey's post hoc comparisons among all nine richness measures.

areas increased both measures on all farms compared to the other more dominant forms of land cover in the region (forest, crops, edges), including large numbers of bees unique to the constructed prairies (Kaiser-Bunbury et al. 2017). These findings were generally consistent among the three farms despite differences in the size and spatial orientation of the constructed habitat. There were significantly lower numbers of bee captures in cover type other than prairie, with edge areas having 35% fewer species than restored prairie, while forest and crops had relatively few species. The latter two cover types appear to play a more limited role in the persistence of native bees on our farms. It should be noted that the deciduous forests we sampled were regenerated from crop fields and pasture over the last half-century, with this forest type, in this region and elsewhere, generally being depauperate in spring floral resources compared to uncut forest (Matlack 1994; MacDougall 2001; Flinn & Vellend 2005; McCune et al. 2017). In total, our work indicates that even small patches of high-quality restored habitat on marginal farm land may support abundant and species-rich bee assemblages compared to other more common and widespread cover types typical of agricultural landscapes. This suggests that prairie restoration may significantly contribute to bee conservation on farms, and potential spill-over effects of increased pollinator services on nearby crops as has been reported elsewhere (Kremen et al. 2004; Carvalheiro et al. 2011; Roulston & Goodell 2011; Morandin et al. 2016; Carvell et al. 2017).

The key factor for most native bee occurrences was floral resources. This applied to community-level measures of total flowers, the total diversity of plant species known to offer nectar and pollen rewards, and the relative abundance of yellow, white, and blue/purple flowers. In contrast, the availability of nesting resources did not strongly predict the richness and abundance of bees. Our surveys suggest that the nesting resources important for bees, from bare ground to herbaceous and woody stems (Potts et al. 2005; Grundel et al. 2010), are available in most of our four cover types, on each farm. Floral resources, in contrast, were more spatially limited to prairie and edges, which may explain why most bees were captured there. These latter results were certainly shaped by the disproportionate capture of a subset of abundant bee genera in prairie, especially *Lasioglossum* which was our most species-rich genera (32 species). Many genera were less strongly associated with prairie, with fewer still more commonly captured in other cover types (e.g. *Andrena*). There could also be some sampling bias, if the timing of our surveys favored foraging activity versus when native bees tend to be seeking nesting sites (Tscharntke et al. 1998). Additionally, the bees may also be nesting in areas outside the farms, or the pan trapping that captured most of our bees may underestimate nesting habitat usage (Sardinas & Kremen 2014).

Overall, we detected mostly generalists with smaller body sizes from the *Augochlorella*, *Agapostemon*, *Andrena*, *Ceratina*, *Hoplitis*, and *Lasioglossum* genera. Most of the rarer species were only found in the prairie. Our capture rates can be considered average or even high compared to other studies (Droege et al. 2010). In terms of nesting guild composition, approximately 70% of the total specimens caught were ground nesters—we had hypothesized that ground nesters might be rare given that undisturbed bare soil can be scarce on agricultural landscapes, but this was not the case. The most obvious explanation is that most of the captured genera—known to be widespread, associated with disturbed habitats, and readily captured by pan trapping (Brosi et al. 2007)—can either

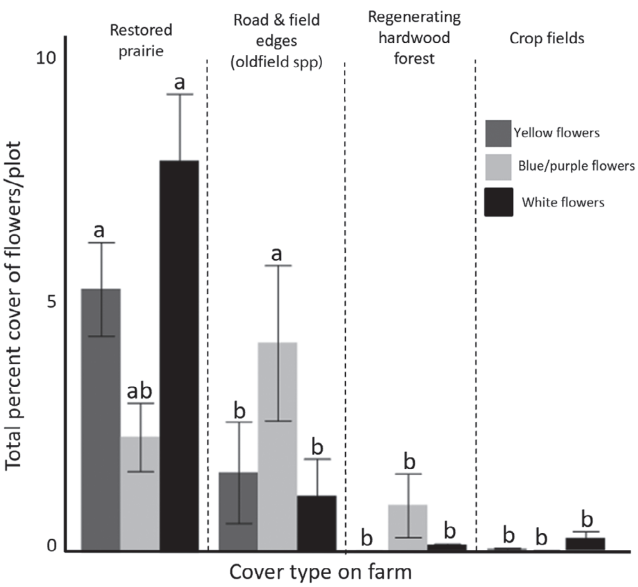

Figure 4. Percent cover of flowers by cover type, measured in one hundred and twenty 16-m^2 plots across three farms. Road and field edges contain non-native pastoral oldfield plant communities including forbs genera such as *Vicia* and *Trifoliium* that can provide floral resources to native bees. Forest patches on the three farms were regenerating from crop fields abandoned at least several decades earlier, and generally lacked flora resources. Floral resources in the crop fields were annual non-native ruderal genera such as *Polygonum* spp. Error bars are ± 1 SE. Letters indicate significant differences among the four cover types, for each flower color separately, derived from Tukey's post hoc comparisons.

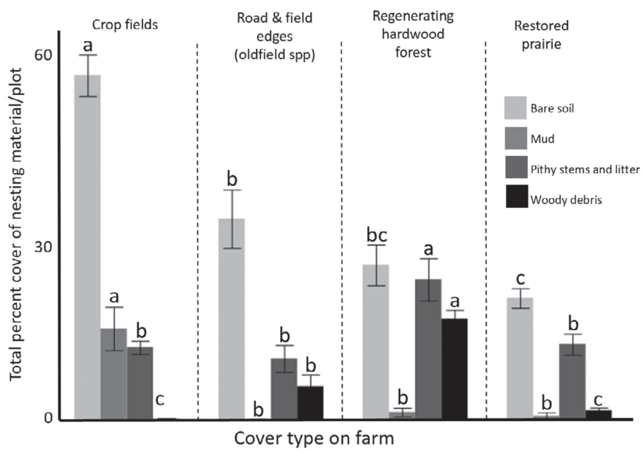

Figure 5. Percent cover of nesting material by cover type, measured in one hundred and twenty 16-m^2 plots across three farms. Error bars are ± 1 SE. Letters indicate significant differences among the four cover types, for each type of nesting material separately, derived from Tukey's post hoc comparisons.

recruit in crop fields despite cultivation or thrive elsewhere including restored prairie, where bare ground was detectable. Twenty-three percent of captured species had nesting life histories associated with pithy stems and twigs, with 3% being cavity nesters, 2% parasitic, and 0.4% wood nesters. There was a steep seasonal decline in the overall abundance and species richness of specimens caught in mid and late summer, despite the significant increase of floral resources in prairie compared to spring. A likely explanation is competition between the flowers and pan traps, with the latter being less effective toward the mid and later summer as floral abundance increased. There is also a possibility that these declines coincide with the periods when agricultural management intensifies including pesticide utilization, but this remains untested.

Given that the planted prairies tended to be less than 10 years old and, on two of the farms, relatively small, it was conceivable that we would detect limited native bee richness with no substantial differences in richness compared other cover types on the farms. This may have been especially likely because the region in which we worked is almost exclusively composed of habitat not assumed to be ideal for pollinators—crops, regenerating forest, and spatially constricted edge areas, with limited quality habitat especially in terms of the summer-long availability of flowers. This was not the case. First, bees appear to be exhibiting long-distance dispersal within the region, such that colonization of new yet spatially isolated high-quality habitat seems to occur relatively rapidly (Greenleaf et al. 2007). Bees targeted flower-rich prairie habitat, apparently regardless of the size and location of these areas. The farm with the largest prairie (Farm 2) showed a much greater spatial separation of bee genera by habitat type, yet the richness, abundance, and composition of the bee community on the other farms with much smaller prairies were not substantially different. We did detect a spatial reduction of pollinators from prairie to crop field, with few individuals captured beyond 50 m from the prairie boundaries—this appears to derive more from a scarcity of floral resources than an inability of bees to reach such areas.

Second, our work shows that large numbers of pollinators occur in edge areas such as field edges and roadsides. Edge areas had some similarities in floral resources with prairie, and supported 16% of all individuals that we captured and 55% of the overall species richness. These results suggest that edges could act as important habitat for native bees in agricultural landscapes, despite their limited extent. Edge areas along roads and fields are typically unmanaged, or if managed done so in ways that are not supportive of pollinators (e.g. plowing or herbicide to control "weeds" or to widen crop fields). As has been seen for the occurrences of milkweed needed by Monarch butterfly (Flockhart et al. 2015), edge habitat can play an important role for supporting communities of native pollinators even if such areas are limited in size. Our results imply that edge habitat in agricultural landscapes could play a similar role for native bee communities—they are limited in area, not necessarily ideal in terms of floral richness and abundance compared to prairie, but nonetheless appear to be highly frequented by many species.

Overall, restored prairie was associated with a much greater abundance and richness of native bees. We did not survey farms with and without restored prairie. However, if we exclude all captures from prairie areas on farms then the remaining edge areas, forest, and crop field (the typical cover profile for farms in our region of North America) would support 33% fewer species, show a 72% reduction in abundance, and fail to detect 35 species that were only seen in prairie. Thus, the relatively small area of prairie within a broad matrix of intensive farming still appears to provide benefit for native pollinators. There is also the need for follow-up work. For example, we could not tell if the high-quality restored areas drew bees away from other cover types where they would otherwise be sampled. Furthemore, the declines over the summer that we observed on all farms could reflect an ecological trap or "extinction debt," where high-quality habitat brings species onto farms that are subsequently adversely affected by management such as pesticide application. These issues point to significant uncertainties concerning the long-term stability of native bee species on agricultural landscapes, including where such species are residing in the absence of resource-rich prairie and whether habitat restoration can meaningfully offset declines of native bees in the long term.

Acknowledgments

Parts of this manuscript have been published in fulfillment of the Master Thesis for Caitlin Paterson, at the University of Guelph. Thanks to field technicians Felicia Syer, Kara Bulger, and Amber Lammers, and the support of ALUS Canada (Bryan Gilvesy, Dave Reid, Casey Whitelock) including the participating farmers who graciously granted access to their farms. Thanks also to Peter Kevan, Steve Marshall, Alex Smith, and their labs for logistical help including species identification. We also acknowledge the comments of two anonymous reviewers, which greatly improved the paper. Funding to C.P. from an OMAFRA HQP Scholarship. Funding to A.S.M. from the Natural Sciences and Engineering Research Council of Canada, the W. Garfield Weston Foundation, and the University of Guelph's Canada First Research Excellence Fund project "Food from Thought."

LITERATURE CITED

Anderson MJ, Crist TO, Chase JM, Vellend M, Inouye BD, Freestone AL, et al. (2011) Navigating the multiple meanings of β diversity: a roadmap for the practicing ecologist. Ecology Letters 14:19–28

Barral MP, Benayas JMR, Meli P, Maceira NO (2015) Quantifying the impacts of ecological restoration on biodiversity and ecosystem services in agroecosystems: a global meta-analysis. Agriculture, Ecosystems & Environment 202:223–231

Biondini M (2007) Plant diversity, production, stability, and susceptibility to invasion in restored northern tall grass prairies. Restoration Ecology 15:77–87

Blaauw BR, Isaacs R (2014) Flower plantings increase wild bee abundance and the pollination services provided to a pollination-dependent crop. Journal of Applied Ecology 51:890–898

Borcard D, Legendre P (2002) All-scale spatial analysis of ecological data by means of principal coordinates of neighbour matrices. Ecological Modelling 153:51–68

Borcard D, Gillet F, Legendre P (2011) Numerical ecology with R. Springer, New York

Brosi BJ, Daily GC, Ehrlich PR (2007) Bee community shifts with landscape context in a tropical countryside. Ecological Applications 17:418–430

Carstensen DW, Sabatino M, Morellato LPC (2016) Modularity, pollination systems, and interaction turnover in plant-pollinator networks across space. Ecology 97:1298–1306

Carvalheiro LG, Veldtman R, Shenkute AG, Tesfay GB, Pirk CWW, Donaldson JS, Nicolson SW (2011) Natural and within-farmland biodiversity enhances crop productivity. Ecology Letters 14:251–259

Carvell C, Meek WR, Pywell RF, Goulson D, Nowakowski M (2007) Comparing the efficacy of agri-environment schemes to enhance bumble bee abundance and diversity on arable field margins. Journal of Applied Ecology 44:29–40

Carvell C, Bourke AF, Dreier S, Freeman SN, Hulmes S, Jordan WC, Heard MS (2017) Bumblebee family lineage survival is enhanced in high-quality landscapes. Nature 543:547–549

Cottenie K (2005) Integrating environmental and spatial processes in ecological community dynamics. Ecology Letters 8:1175–1182

Cox-Foster DL, Conlan S, Holmes EC, Palacios G, Evans JD, Moran NA, et al. (2007) A metagenomic survey of microbes in honey bee colony collapse disorder. Science 318:283–287

Droege S, Tepedino VJ, Lebuhn G, Link W, Minckley RL, Chen Q, Conrad C (2010) Spatial patterns of bee captures in North American bowl trapping surveys. Insect Conservation and Diversity 3:15–23

Flinn KM, Vellend M (2005) Recovery of forest plant communities in post-agricultural landscapes. Frontiers in Ecology and Environment 3:243–250

Flockhart DT, Pichancourt JB, Norris DR, Martin TG (2015) Unravelling the annual cycle in a migratory animal: breeding-season habitat loss drives population declines of monarch butterflies. Journal of Animal Ecology 84:155–165

Garibaldi LA, Steffan-Dewenter I, Winfree R, Aizen MA, Bommarco R, Cunningham SA, Bartomeus I (2013) Wild pollinators enhance fruit set of crops regardless of honey bee abundance. Science 339:1608–1611

Germain R, Johnson L, Schneider S, Cottenie K, Gillis L, MacDougall AS (2013) Spatial variability in plant predation determines the strength of stochastic community assembly. American Naturalist 182:169–179

Greenleaf S, Williams N, Winfree R, Kremen C (2007) Bee foraging ranges and their relationships to body size. Oecologia 153:589–596

Grundel R, Jean RP, Frohnapple KJ, Glowacki GA, Scott PE, Pavlovic NB (2010) Floral and nesting resources, habitat structure, and fire influence bee distribution across an open-forest gradient. Ecological Applications 20:1678–1692

Hannon LE, Sisk TD (2009) Hedgerows in an agri-natural landscape: potential habitat value for native bees. Biological Conservation 142:2140–2154

Hines HM, Hendrix SD (2005) Bumble bee (Hymenoptera: Apidae) diversity and abundance in tallgrass prairie patches: effects of local and landscape floral resources. Environmental Entomology 34:1477–1484

Holzschuh A, Steffan-Dewenter I, Kleijn D, Tscharntke T (2007) Diversity of flower-visiting bees in cereal fields: effects of farming system, landscape composition and regional context. Journal of Applied Ecology 44:41–49

Hopwood JL (2008) The contribution of roadside grassland restorations to native bee conservation. Biological Conservation 141:2632–2640

Jack BK, Kousky C, Sims KR (2008) Designing payments for ecosystem services: lessons from previous experience with incentive-based mechanisms. Proceedings of the National Academy of Sciences of the United States of America 105:9465–9470

Kaiser-Bunbury CN, Mougal J, Whittington AE, Valentin T, Gabriel R, Olesen JM, Blüthgen N (2017) Ecosystem restoration strengthens pollination network resilience and function. Nature 542:223–227

Kennedy CM, Lonsdorf E, Neel MC, Williams NM, Ricketts TH, Winfree R, et al. (2013) A global quantitative synthesis of local and landscape effects on wild bee pollinators in agroecosystems. Ecology Letters 16:584–599

Kerr JT, Deguise I (2004) Habitat loss and the limits to endangered species recovery. Ecology Letters 7:1163–1169

Kim J, Williams N, Kremen C (2006) Effects of cultivation and proximity to natural habitat on ground-nesting native bees in California sunflower fields. Journal of the Kansas Entomological Society 79:309–320

Koh I, Lonsdorf EV, Williams NM, Brittain C, Isaacs R, Gibbs J, Ricketts TH (2016) Modeling the status, trends, and impacts of wild bee abundance in the United States. Proceedings of the National Academy of Sciences of the United States of America 113:140–145

Kovács-Hostyánszki A, Espíndola A, Vanbergen AJ, Settele J, Kremen C, Dicks LV (2017) Ecological intensification to mitigate impacts of conventional intensive land use on pollinators and pollination. Ecology Letters 20:673–689

Krebs CJ (1998) Ecological methodology. Benjamin-Cummings Press, San Francisco, California

Kremen C, Williams NM, Thorp RW (2002) Crop pollination from native bees at risk from agricultural intensification. Proceedings of the National Academy of Sciences of the United States of America 99:16812–16816

Kremen C, Williams NM, Bugg RL, Fay JP, Thorp RW (2004) The area requirements of an ecosystem service: crop pollination by native bee communities in California. Ecology Letters 7:1109–1119

MacDougall AS (2001) Conservation status of Saint John River valley hardwood forest in western New Brunswick. Rhodora 103:47–70

MacDougall AS, Wilson SD, Bakker JD (2008) Climatic variability alters the outcome of long-term community assembly. Journal of Ecology 96:346–354

Marshall SA (2007) Insects: their natural history and diversity. Firefly Books, Buffalo, NY

Matlack GR (1994) Plant species migration in a mixed-history forest landscape in eastern North America. Ecology 75:1491–1502

McCune JL, Van Natto A, MacDougall AS (2017) The efficacy of protected areas and private land for plant conservation in a fragmented landscape. Landscape Ecology 32:871–882

Morandin LA, Kremen C (2013a) Bee preference for native versus exotic plants in restored agricultural hedgerows. Restoration Ecology 21:26–32

Morandin LA, Kremen C (2013b) Hedgerow restoration promotes pollinator populations and exports native bees to adjacent fields. Ecological Applications 23:829–839

Morandin LA, Winston ML (2005) Wild bee abundance and seed production in conventional, organic, and genetically modified canola. Ecological Applications 15:871–881

Morandin LA, Long RF, Kremen C (2016) Pest control and pollination cost–benefit analysis of hedgerow restoration in a simplified agricultural landscape. Journal of Economic Entomology 109:1020–1027

Öckinger E, Smith HG (2007) Semi-natural grasslands as population sources for pollinating insects in agricultural landscapes. Journal of Applied Ecology 44:50–59

Packer L, Genaro JA, Sheffield CS (2007) The bee genera of eastern Canada. Canadian Journal of Arthropod Identification 3:1–32

Paterson C (2014) Small restoration, big impacts: how habitat influences native pollinators in intensive agricultural landscapes. MSc thesis. Department of Integrative Biology, University of Guelph, Guelph, Canada

Pinto SM, MacDougall AS (2010) Dispersal limitation and environmental structure interact to restrict the occupation of optimal habitat. American Naturalist 175:675–686

Ponisio LC, M'gonigle LK, Kremen C (2016) On-farm habitat restoration counters biotic homogenization in intensively managed agriculture. Global Change Biology 22:704–715

Potts SG, Vulliamy B, Roberts S, O'Toole C, Dafni A, Ne'eman G, Willmer P (2005) Role of nesting resources in organising diverse bee communities in a Mediterranean landscape. Ecological Entomology 30:78–85

Riley J (2013) The once and future Great Lakes country: an ecological history. McGill-Queens University Press, Kingston, Ontario, Canada

Robertson GP, Swinton SM (2005) Reconciling agricultural productivity and environmental integrity: a grand challenge for agriculture. Frontiers in Ecology and Environment 3:38–46

Roulston TAH, Goodell K (2011) The role of resources and risks in regulating wild bee populations. Annual Review of Entomology 56:293–312

Sardinas HS, Kremen C (2014) Evaluating nesting microhabitat for ground-nesting bees using emergence traps. Basic and Applied Ecology 15:161–168

Scheper J, Bommarco R, Holzschuh A, Potts SG, Riedinger V, Roberts SP, Wickens VJ (2015) Local and landscape-level floral resources explain effects of wildflower strips on wild bees across four European countries. Journal of Applied Ecology 52:1165–1175

Schuler RE, Roulston TH, Farris GE (2005) Farming practices influence wild pollinator populations on squash and pumpkin. Journal of Economic Entomology 98:790–795

Spafford RD, Lortie CJ (2013) Sweeping beauty: is grassland arthropod community composition effectively estimated by sweep netting? Ecology and Evolution 3:3347–3358

The Xerces Society (2011) Attracting native pollinators: protecting North America's bees and butterflies. Storey Publishing, North Adams, Massachusetts

Tscharntke T, Gathmann A, Steffan-Dewenter I (1998) Bioindication using trap-nesting bees and wasps and their natural enemies: community structure and interactions. Journal of Applied Ecology 35:708–719

Williams NM, Ward KL, Pope N, Isaacs R, Wilson J, May EA, et al. (2015) Native wildflower plantings support wild bee abundance and diversity in agricultural landscapes across the United States. Ecological Applications 25:2119–2131

Winfree R (2010) The conservation and restoration of wild bees. Annals of the New York Academy of Sciences 1195:169–197

Winfree R, Williams NM, Dushoff J, Kremen C (2007) Native bees provide insurance against ongoing honey bee losses. Ecology Letters 10:1105–1113

Winfree R, Aguilar R, Vázquez DP, LeBuhn G, Aizen MA (2009) A meta-analysis of bees' responses to anthropogenic disturbance. Ecology 90:2068–2076

Wood TJ, Holland JM, Goulson D (2015) Pollinator-friendly management does not increase the diversity of farmland bees and wasps. Biological Conservation 187:120–126

Supporting Information

The following information may be found in the online version of this article:

Table S1. List of species trapped on the three farms (2012). Data are pooled over three sampling periods (June, July, August).
Table S2. Species sampled in low numbers, with frequency among the three farms, the habitat where the species was sampled (mostly prairie and edge), and the nesting type for which the species is associated (The Xerces Society 2011).
Table S3. Summary measures of sampled bees per habitat type, pooled over three sampling periods.
Table S4. Correlation matrix among the measured variables for Farm 1, which was composed of mostly corn fields, with small amounts of edge habitat especially along roadsides, regenerating forest, and a small restored prairie (see Fig. S1).
Table S5. Correlation matrix among the measured variables for Farm 2, which was composed of mostly restored prairie and oak savanna, with some regenerating forest, edge areas, and orchards (see Fig. S1).
Table S6. Correlation matrix among the measured variables for Farm 3, which was composed of mostly soy and corn fields, with small amounts of edge habitat especially along roadsides, regenerating forest, and a small restored prairie (see Fig. S1).
Figure S1. For each farm, (A) sampling distribution of plots and (B) PCNM (Principal Coordinates of Neighbourhood Matrix) eigenvectors at two spatial levels (see Appendix S1) for species-level analysis of the degree of compositional similarity by location.
Figure S2. Species accumulation curves for the four habitat types, on the three farms.
Appendix S1. Supplementary Methods

Coordinating Editor: Stephen Murphy

Received: 15 November, 2018; First decision: 2 January, 2019; Revised: 23 May, 2019; Accepted: 23 May, 2019; First published online: 27 June, 2019